



BrightLearn Data Analytics

Research Assignment 1

Foundations of Data Analytics & Data Science

Course	Data Analytics & Data Science
Assignment No.	Research Assignment 1
Target Audience	Beginner Students
Due Date	To be confirmed by instructor
Format	Handwritten answers — scan and email your completed work

Instructions

Read each question carefully and research the answer before writing.

- All answers must be handwritten — typed submissions will not be accepted.
- Write in full sentences and use your own words.
- Include diagrams or examples where relevant — this will earn you bonus marks.
- Cite your sources at the bottom of each section.
- Scan your completed assignment and submit it before the due date.

Section A: Data Fundamentals



Research Tip: Start with official documentation from platforms like IBM, Google Cloud, AWS, or Microsoft Azure — they offer excellent beginner-friendly explanations.

1. What is a Database? Define it in your own words and explain why databases are essential in the modern world. Provide two real-world examples of where databases are used.

A database is a collection of data stored electronically so that it can be easily accessed, managed and





updated. Databases are essential because they allow organisations to store huge amount of information reliably and retrieve what they need fast.

E.g. University uses database to store students information such as marks etc.
Hospital use database to store patients information.

2. What is a Data Warehouse? How does it differ from a regular database? Why would a company use a data warehouse?

is a centralised repository that stores integrated, historical data from Operational systems, structured for analytical queries rather than transactional processing. A database is optimized for fast, frequent read/write operations for one application where-as a data warehouse is optimized for complex queries across historical data from many sources.

- companies use data warehouse to get a consistent view of analysis of the business for decision making.

3. What is a Data Lake? Explain the concept and describe how it differs from a data warehouse. When would a business prefer a data lake over a data warehouse?

It is a storage repository that holds raw data, structured, semi-structured, and unstructured format, without needing to organize it first. Data lake differ from data warehouse because it consist of messy data, it is a messy storage bin, anything can go in immediately, no prep needed, but it takes more work to find and make sense of things later. Business prefer a lake when they have huge volume of varied data and dont yet know exactly how it will be analyzed.

4. What is a Data Mart? How does it relate to a data warehouse and who typically uses it?

is a smaller, more focused subset of a data warehouse, build around the needs of a single department or business unit/function.

It relate to data warehouse by pulling a relevant part of data warehouse. It is typically used by specific teams who needs fast, simple access to just their own data without having to search entire warehouse

5. Explain the difference between Structured, Semi-Structured, and Unstructured Data. Provide one example of each.

- **Structured data**: Follows a strict, predefined format
Example: Class register, every entry follows the exact format

- **Semi-structured data**: It does not neatly fit into tables but has some organisational tags or markers. Example: Email, it has structured fields like sender and subjects but unstructured body.

- **Unstructured data**: has no predefined format at all.

Example: a customer free-text review.





6. What is Data Modeling? Why is it important when designing a system that works with data? Describe at least two types of data models.
 is the process of designing how data will be organized, structured, and related within a system before its build. It is important because a good data model prevent duplication, keeps data consistent, and make the system easier to query and scale.

2 Types of data model:

- Conceptual model: a high-level view of the main entities and their relationship eg customer place an order.
- Relational model: defines specific tables, columns, data type & keys.

7. What is Data Mining? How is it used in business? Provide two real-world examples of data mining in action.
 is the use of data to spot patterns that helps predict behavior or make smarter decisions. Business use data mining to make better decisions.

Example: Bank go through an individual spending pattern and notices sudden transaction that is unusual then flag it as possible fraud.

Checkers notice that people who buy comflakes they also buy milk, they put this two as combo on special.

8. What is a Database Management System (DBMS)? Name and briefly describe four popular DBMS platforms (e.g., MySQL, PostgreSQL, Oracle, MongoDB) and explain a suitable use case for each.

is a software that lets users create, read, update, delete and manage data in a database.

- MySQL: a free, widely used relational DBMS, great for web application
- PostgreSQL: it is an advanced open-source operational DBMS known for handling complex queries and datatypes
- Oracle database: commercial DBMS used heavily by large enterprises for mission-critical systems.
- MongoDB: a noSQL ~~that~~ DBMS that store flexible, JSON-like data: good for applications with rapidly changing or unstructured data

9. What is ETL (Extract, Transform, Load)? Explain each step and describe why this process is important in data analytics.
 is a process of getting messy data from different places by ~~putting~~ cleaning it up and make it consistent, put the cleaned data into its final destination, like a data warehouse.

Extract: pull raw data out of storage.

Transform: clean it up and make it consistent

Load: put cleaned data into its final destination

~~This~~ This process is important because without it the organisations would be comparing messy, inconsistent data across the business. ETL ensures that analysts are working with clean, stable, reliable and unified data.





10. What are common data storage formats used in analytics (e.g., CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another.

1. CSV (Comma-separated values): a spreadsheet saved as a text. plain text, values separated by commas.
 - Used when data is small and simple, and you just need something anyone can be able to use or open.
2. XML (extensible mark-up language): an older tag-based format, similar in spirit to JSON but more verbose (use opening/closing tags)
 - Used when working with older enterprise system or industries that rely on strict, tag-based document standard
3. JSON (javascript object notation): Stores data as labeled key-value pairs and can hold nested, grouped information.
 - Used when data isn't a clean table, like a customer record that has a list of multiple orders inside it.
4. Avro: is a compact binary format that bundles the data's structure together with the data itself.
 - Used when you're streaming data continuously and the data structure might change over time
5. Parquet: a compressed format that stores data by column instead of by row, ~~built~~ built for fast ~~anal~~ analysis of huge datasets.
 - Use when you're doing big-data analysis and only need certain columns out of a massive dataset.





Section B: Artificial Intelligence & Machine Learning

Research Tip: Explore resources like Google's Machine Learning Crash Course, IBM's AI explanations, or OpenAI's blog to build a strong foundational understanding.

11. What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (Weak AI) and General AI (Strong AI). Provide one example of each. AI is a method of getting computers to do things that normally need a human brain, like understanding speech, recognizing pictures or making decisions.

Narrow AI: refers to systems that designed to perform one specific task well, but cannot generalize beyond that task. e.g. Siri answering voice

General AI: refers to a hypothetical system that could understand, learn, and apply intelligence across any task at a human level. e.g. a machine that could do anything a human can / Claude, Co-pilot

12. What is Machine Learning (ML)? How does it differ from traditional programming? Describe the three main types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.

ML is a branch of AI that focuses on developing models, algorithms that let computers learn from data without being explicitly programmed for every task. Traditional programming developer writes exact rules for how to process input to output but in ML, the system is fed with examples of data and learn the rules itself.

Supervised: the model is trained on labeled data

Unsupervised: the model works with unlabeled data and tries to find hidden patterns or grouping on its own.

Reinforcement learning: learns by taking actions in an environment and receiving rewards or penalties, gradually improving its strategy.

13. What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how do they work? IS a specialised subset of machine learning that uses layered structured called neural networks

It relates to ML and AI as, AI is the broadest field, ML is a way of achieving AI, and deep learning is a particular approach within ML that uses many layered networks

Neural networks: work by passing data through multiple layers of interconnected nodes, each node applies a mathematical transformation and the network adjust the strength of connections during training so that its output increasingly match correct answers.

14. What is Natural Language Processing (NLP)? Describe what it is and provide three real-world applications of NLP that you use in everyday life.

IS an AI focused on enabling computers to understand, interpret, and generate human language, whether spoken or written.

Examples:

Voice assistants like Siri, Spam filters analysing email text to detect junk mail, Predictive text / autocorrect suggesting next word as you type on a phone.





15. What is a Large Language Model (LLM)? How is it different from a standard Machine Learning model? Name two well-known LLMs and explain what they are used for.
 is a learning model trained on massive amounts of text data to understand and generate human-like language. Standard ML is trained for one narrow task while LLM is trained on enormous, diverse text dataset and can perform many different language tasks using same underlying model.

GPT: powers ChatGPT, used for chatting, writing and answering questions.
Claude: used for writing help, coding help, analyzing text.

16. What is Generative AI? How does it work at a high level? Name three tools that use Generative AI and describe what each one does.
 is an AI system that can create new content, text, images, audio, video or code rather than ~~analyzing~~ analysing or classifying existing data. At a high level this models learn the underlying patterns and structures of huge training datasets, then use that learned knowledge to generate new, original outputs that follow similar patterns. Tools: ChatGPT / Claude - generate written text and answer questions. Midjourney - generate images from text descriptions. GitHub Copilot - generate programming code suggestions based on context.

17. What is a Training Dataset? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect?
 is a set / collection of data used to teach a machine learning model to recognize patterns and make predictions. Quality of training data is important because model can only learn what is present in the data it's shown, if data is incomplete, inaccurate, or unrepresentative, the model's output will be unreliable. If training data is biased or incorrect, the resulting model may perform poorly or unfairly.

18. What is Computer Vision? How do machines 'see' and interpret images? Provide two real-world use cases of computer vision.

Computer vision is a field in AI that enables machine to interpret and understand visual information from images or video, similar to how human vision works.





Machine "See" by converting images or videos into numerical data then using trained models, such as deep neural networks to detect patterns such as edges, shapes and objects within that data.

Two real-world use cases of Computer Vision

- facial recognition : used to unlock smartphones, ~~and quality control~~
- motion detector cameras / light : use to switch on if a motion of object is detected.





Section C: APIs, Tools & Developer Concepts

Research Tip: Search for 'What is an API for beginners' on YouTube or read documentation from platforms like Postman, RapidAPI, or Anthropic.

19. What is an API (Application Programming Interface)? Explain what it is using a simple real-world analogy. Why are APIs important in modern software and data systems?
 is a set of rules that allows one piece of software to communicate with another, without each needing to know the other's internal workings.
 Example: Vending machine - you press a button (make a request) and machine gives you a snack (response).

API are important because they let different applications, services, and devices share data and functionality efficiently enabling everything from weather apps pulling live forecasts to payment processing.

20. What is an API Key? What is its purpose? How should it be stored and protected, and what could happen if it is exposed publicly?
 is a unique code that identifies and authenticates the application or user making a request to an API. Its purpose is to track usage, control access, and enforce rate limits, ensuring only authorized users can access the service. It should be stored securely, never hard-coded directly into public source code, using environment variables or secret-management tools. If API key is exposed publicly, unauthorized users could use it to make request on owner's account and accessing sensitive data.

21. What is a REST API? What does REST stand for and what are the main HTTP methods (GET, POST, PUT, DELETE) used in REST APIs? Briefly describe each method.

Representational State Transfer (REST) - is an architectural style of designing web APIs that uses standard HTTP method to interact with resources identified by URLs.

GET: retrieves data from the server without changing anything.

POST: sends new data to the server to create a new resource.

PUT: updates an existing resource, typically replacing it entirely with new data provided.

DELETE: removes specified resource from the server.

22. What is JSON (JavaScript Object Notation)? Why is it commonly used when working with APIs and data exchange? Write a simple example of a JSON object.
 It is a lightweight, text-based format for storing and exchanging data, structured as key-value pairs and nested objects/arrays.

It is commonly used when working with APIs because its easy for both humans to read and machines





to parse and it maps naturally onto data structures used in most programming languages.

Example: `{ "name": "pinch", "isStudent": true, "institution": "brightlearn" }`

23. What is Prompt Engineering? Why is it an important skill when working with AI tools like ChatGPT or Claude? Provide two examples of a poorly written prompt and an improved version of each.

is a practice of carefully crafting the instructions/prompts given to an AI to get the most accurate, useful, relevant response.

It is an important skill when working with tools like ChatGPT or Claude because they answer/respond directly to how a question or instruction is phrased.

Example of poor prompt: "help me with resume"
Improved version: "Review my resume for a ~~marketing~~ analyst job and suggest specific improvement to make my experience section stronger"

Poor: "Explain AI", Improved version: "Explain AI for a beginner using everyday analogy"

24. What is a MAP function? Explain what it does in programming and data processing. Provide a simple example in either Python or JavaScript. MAP take a list of items, does the same action to every single item and gives you back a new list with results.

Example in Python:

```
Numbers = [1, 2, 3, 4]
```

```
Squared = list(map(lambda x: x**2, numbers))
```

```
# Squared = [1, 4, 9, 16]
```

25. What is Cloud Computing? Name the three main cloud service models (IaaS, PaaS, SaaS) and briefly explain each. Give one example of a product for each model. is a delivery of computing services, servers, storage, database, software over the internet rather than relying on local hardware.

IaaS: provides raw computing infrastructure like virtual servers and storage that users manage themselves.

PaaS (Platform as service): provides ready made platforms for developers to build and deploy applications without managing the underlying infrastructure.

SaaS (Software as service): provides fully built, ready to use software applications accessed over the internet. Examples of this service models: Amazon EC2, Google App engine, gmail.

26. What is a Version Control System? What is Git and why is it widely used by data analysts and developers? Briefly explain what a 'commit' and a 'branch' are.

is a tool that tracks changes to files over time, allowing multiple people to collaborate on the same project while keeping a complete history of edits that can be reviewed or reverted

Git is the most widely used VCS, because it allows safe experimentation





and collaboration and rollback mistakes among data analysts & developers.
Commit - is a saved snapshot of changes made to the project
Branch - is separate, parallel version of the project where changes can be made independently before being merged back into main codebase.
27. What is a Jupyter Notebook? What programming language is it most commonly used with? Why is it popular among Data Analysts and Data Scientists?

Jupyter notebook is an interactive tool, web-based tool that allows users to write and run code in small sections, alongside text, visualizations and equations all within a single document. It is commonly used with Python. It is popular because it allows them to explore data, test code incrementally, view results and charts immediately and document their reasoning in one place.

28. What is Python? Why is it considered one of the most popular programming languages for Data Analytics and AI? Name four Python libraries commonly used in data work and briefly describe each.

Python is a high-level, general purpose programming language known for its simple, readable syntax.

It is considered most popular programming language for data analytics and AI because its relatively easy to learn, has a huge community and offers a rich ecosystem of specialized libraries.

Four Python libraries used in data work:

- Pandas: used for organizing, cleaning, and analyzing tabular data.
- Numpy: provides support for fast numerical computations and array operations.
- Matplotlib: used for creating charts and data visualization
- Scikit learn: provides tools for building and evaluating machine learning models.





Section D: Data Ethics, Governance & Careers

Research Tip: Explore the World Economic Forum, GDPR documentation, and LinkedIn Learning for content on data ethics and career pathways in data.

29. What is Data Privacy? Why is it important? Research the GDPR (General Data Protection Regulation) and explain what it is and which countries it applies to. Data privacy refers to the right practice of ensuring that individuals' personal information is collected, stored, and used appropriately, with their knowledge and consent. It is important because personal data such as financial details, health records or browsing habits can cause harm if misused. GDPR - is a data protection law that sets strict rules on how organizations collect, store and process personal data, giving individual right such as access to data. It applies in European Union.
30. What is Data Ethics? Provide three examples of unethical use of data or AI, and explain what makes each example problematic. Data ethics involves the moral principles guiding how data is collected, used and shared responsibly.
- Example of unethical use of data or AI
- * Using facial recognition technology to track individuals without their consent, which violates privacy.
 - * Training a hiring algorithm on historical data that reflects past discrimination, causing it to unfairly reject candidates from certain groups.
 - * Selling users' personal information/data to third parties without informing them.
31. What is Data Governance? Define it and explain the key components of a good data governance framework. Data governance is a framework of policies, processes, and responsibilities that ensure an organisation's data is accurate, secure, consistent and used appropriately. Key components of a good data governance framework include: clear data ownership and accountability, data quality standards, security and access controls, compliance with relevant laws and regulations, and documented process for how data is collected, stored & used.
32. What is Bias in AI? Explain what it means for an AI model to be biased. Provide a real-world example of AI bias and describe the potential consequences. Bias AI occurs when a model ~~produce~~ produces systematically unfair or skewed outcomes, often because the data it was trained on reflects existing imbalances or prejudices in the real world.

e.g. AI tool trained mostly on resumes from male applicants that learns to favor male candidates over equally qualified female candidates





Consequences include unfair treatment of certain groups, reinforcement of societal inequalities, and legal or reputational risk for organisation using the biased system.

33. What are the key differences between a Data Analyst, a Data Scientist, a Data Engineer, and an AI Engineer? Create a simple table comparing their roles, skills, and tools.

Role	Skills	Tools
Data analyst: interpret existing data to answer business questions and create reports	SQL, statistics, data visualisation	Excel, SQL, Power BI, Tableau
Data Scientist: builds predictive models and uncover deeper insights from data	Statistics, programming, machine learning	Python/R, Jupyter, Scikit-learn
Data engineer: builds and maintain the infrastructure / pipelines that move and store data	database design, ETL processes, cloud system	SQL, Spark, cloud platforms (AWS, Azure)
AI engineer: design, build, and deploys AI/ML system into production application	ML, software engineering, deep learning frameworks	Python, TensorFlow, Cloud AI Services

34. What is Business Intelligence (BI)? Name two popular BI tools (e.g., Power BI, Tableau) and explain how they help businesses make data-driven decisions.

BI refers to the technologies and processes used to collect, analyze and present business data in a way that supports informed decision-making.

Power BI: Connects to various data sources and creates interactive dashboards and reports.

Tableau: It is known for strong data visualization capabilities. This tool help business by turning raw data, complex data into clear visual dashboards that highlights trends, outliers and performance indicators

Section E: Personal Reflection

Answer the following reflection questions thoughtfully and honestly. There is no right or wrong answer — we want to know what you think!

35. Which concept from this assignment did you find most interesting or surprising? Explain why.

Section A. It made me understand and picture how data works in the system.

~~Section A~~ Section C: It gave me more clarity of API and answered some of questions I had about API.

36. Which concept did you find most difficult to understand? What resources did you use to help you understand it better?

Section BD, I used AI tool called Claude to

explain in simplest form.





37. How do you see Data and AI affecting your future career or everyday life? Be specific with examples.

Tools are increasingly used to analyze requirements documents test cases are created with tools, this has been happening manually, AI speed up the design of test case and as a tester I can focus on higher risk tasks as well as doing the checks on whether all is well.

References

List all the sources you used to complete this assignment. Use the format below:

Author/Organisation. (Year). Title of article or webpage. Website Name. URL

1. IBM (2026). What is a database? IBM. <https://www.ibm.com/think/topics/databases>
2. Google cloud. (2025). What is data warehouse? google cloud <https://cloud.google.com/learn/what-is-datawarehouse>
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6. IBM content team (2026). What is database schema? IBM. <https://www.ibm.com/think/topics/schema>
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8. Stonier, J., & Woodman, L. (2024). Why data equity plays a key role in creating trust in tech. world economic forum. <https://www.weforum.org/stories/2024/01/data-equity-trust/>

Good luck! We believe in you.
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